

Problem

The circuit in Fig. 8.123 is the electrical analog of body functions used in medical schools to study convulsions. The analog is as follows:

C_1 = Volume of fluid in a drug

C_2 = Volume of blood stream in a specified region

R_1 = Resistance in the passage of the drug from the input to the blood stream

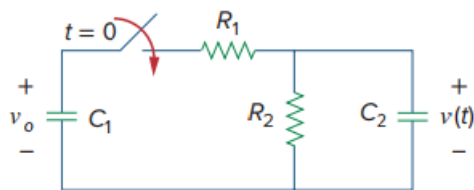
R_2 = Resistance of the excretion mechanism, such as kidney, etc.

v_0 = Initial concentration of the drug dosage

$v(t)$ = Percentage of the drug in the blood stream

Find $v(t)$ for $t > 0$ given that $C_1 = 0.5 \mu\text{F}$, $C_2 = 5 \mu\text{F}$, $R_1 = 5 \text{ M}\Omega$, $R_2 = 2.5 \text{ M}\Omega$, and $v_0 = 60u(t)$ V.

Figure 8.123:

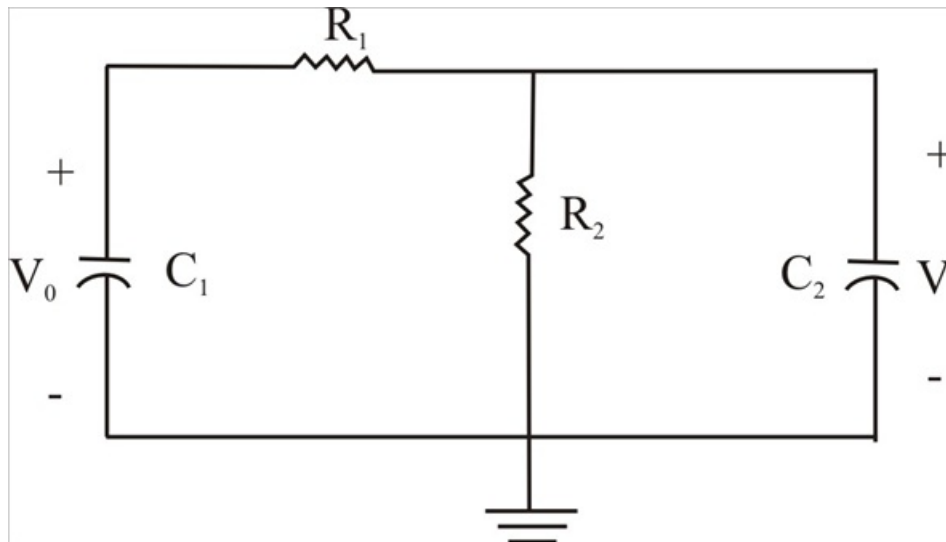


Step-by-step solution

Step 1 of 2

For $t = 0^-$, $V(0) = 0$.

For $t > 0$, Circuit is as shown below.



Step 2 of 2

At node a,

$$\left(V_0 - \frac{V}{R_1} \right) = \frac{V}{R_2} + C_2 \frac{dV}{dt}$$

$$V_0 = V \left(1 + \frac{R_1}{R_2} \right) + R_2 C_2 \frac{dV}{dt}$$

$$60 = \left(1 + \frac{5}{2.5} \right) V + (5 \times 10^6 \times 5 \times 10^{-6}) \frac{dV}{dt}$$

$$60 = 3V + 25 \frac{dV}{dt}$$

$$V(t) = V_s + \left(A e^{\frac{-3t}{25}} \right)$$

$$3V_s = 60 \text{ yields } V_s = 20$$

$$V(0) = 0 = 20 + A, \quad A = -20$$

$$V(t) = 20 \left(1 - e^{\frac{-3t}{25}} \right)$$